

t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE is a **dimensionality reduction and visualization technique**.

Its main goal is:

"Take high-dimensional data and display it in 2D or 3D while keeping similar points close together."

Unlike PCA, t-SNE is primarily used for **visualization**, not feature reduction for model training.

Why Do We Need t-SNE?

Imagine you have customer data with:

```
Age
Income
Location
Spending Score
Website Visits
Purchases
...
50 features
```

Humans cannot visualize 50 dimensions.

t-SNE converts:

```
50 dimensions
      ↓
2 dimensions
```

so we can plot the data.

Real-Life Analogy

Imagine 100 students in a school.

Some are:

- Cricket lovers
- Football lovers
- Music lovers
- Gamers

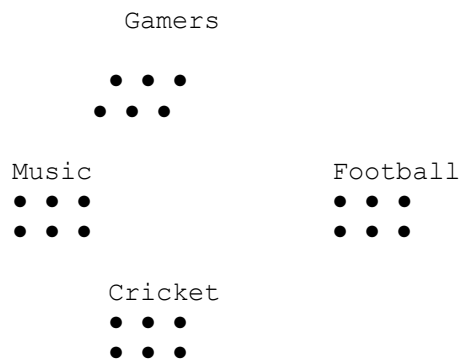
Each student has 50 characteristics.

You can't visualize 50 characteristics.

t-SNE tries to place students on a 2D map such that:

Friends stay close together
Different groups stay apart

Result:



The exact distances are not important.

What matters is:

Similar points stay together.

How PCA and t-SNE Differ

Suppose you have 3 groups:

Students
Employees
Retired People

PCA

PCA preserves overall variance.

You may get:



Clusters overlap.

t-SNE

t-SNE preserves neighborhoods.

You may get:

Cluster A	Cluster B	Cluster C
•••••	•••••	•••••
•••••	•••••	•••••

Much easier to visualize.

Simple Example

Suppose we have these people:

Person	Height	Weight	Age
A	170	65	20
B	171	66	21
C	169	64	19
D	190	95	40
E	191	96	41
F	189	94	39

Clearly:

Group 1:
A B C

Group 2:
D E F

In 3D space:

Height
Weight
Age

These groups are separated.

t-SNE projects them into 2D:

A B C

D E F

The grouping remains visible.

The Main Idea

Instead of preserving actual distances, t-SNE preserves:

Who is close to whom?

For example:

Original space:

A close to B
A close to C

A far from D
A far from E

After projection:

A B C stay together

D E F stay together

This is what t-SNE tries to maintain.

How It Works (Intuition)

Step 1

Measure similarity in high dimensions.

Example:

A ↔ B = very similar
A ↔ C = similar
A ↔ D = not similar

Convert similarities into probabilities.

Step 2

Create random 2D positions.

Initially:

A
D
B
F
C
E

all mixed up.

Step 3

Iteratively move points.

The algorithm repeatedly adjusts positions:

Similar points attract
Dissimilar points repel

until neighborhood relationships are preserved.

Customer Segmentation Example

Suppose an e-commerce company has:

100,000 customers

Each customer has:

Age
Income
Orders
Returns
Visits
Cart Value
...
50 features

After t-SNE:

Budget Buyers

●●●●●

.....

Premium Buyers

.....
.....

Impulse Shoppers

.....
.....

Business teams can immediately see customer segments.

t-SNE in AI

Very common in:

Image Classification

Suppose a neural network produces:

512 features per image

t-SNE visualizes them:

Cats cluster together

Dogs cluster together

Cars cluster together

This helps verify that the model learned meaningful representations.

Single-Cell Genomics

One of the most famous use cases.

Researchers may have:

20,000 gene-expression features

per cell.

t-SNE reduces:

20,000
↓
2

allowing different cell types to appear as separate clusters.

This is why many biology papers show colorful t-SNE plots.

Important Limitation

Many beginners make this mistake:

Wrong Interpretation

If Cluster A and B are twice as far apart as Cluster A and C:

A ---- C ----- B

you cannot conclude:

"B is twice as different as C."

t-SNE does **not** preserve global distances well.

It mainly preserves local neighborhoods.

PCA vs t-SNE

Feature	PCA	t-SNE
Goal	Reduce dimensions	Visualize clusters
Speed	Fast	Slow
Linear	Yes	No
Preserves variance	Yes	No
Preserves neighborhoods	Limited	Excellent
Model preprocessing	Common	Rare
Visualization	Good	Excellent

Typical Workflow

In practice:

100 Features
↓
PCA (100 → 30)
↓
t-SNE (30 → 2)
↓
Visualization

This is extremely common because PCA removes noise and speeds up t-SNE.

Interview Question

Why is t-SNE preferred over PCA for visualization?

Answer:

PCA preserves variance, whereas t-SNE preserves local neighborhood structure. Therefore, t-SNE often produces clearer visual separation between clusters and is better suited for exploring hidden patterns in high-dimensional data.

Memory Trick

Imagine organizing people at a party:

- PCA asks:
"Which direction captures the most variation?"
- t-SNE asks:
"Who are friends, and how can I keep friends together?"

So:

PCA = Preserve variance

t-SNE = Preserve neighbors

That's the easiest way to remember the difference.